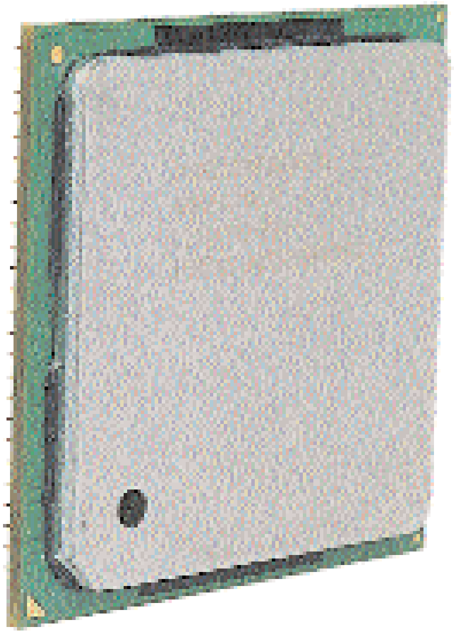
Communication unlimited

Cellular technology already has a strong foothold in our daily lives. While speech is still the main medium for cellular networks, many upcoming technologies are built to open the doors to newer forms of media such as audio and video broadcasts, videoconferencing and advanced messaging systems. While devices that support these technologies are still beyond the reach of most users, it is just a matter of time before this technology becomes widely accessible.

GPRS is among the most significant cellular technologies this year. It is gaining acceptance among service providers as well as end users. With the ability to handle data transmission at a maximum of 144 Kbps, this technology is a taste of the potential of a handled device to deliver more than just voice communication. This year, expect to see the proliferation of GPRS-based networks along with the advent of newer standards such as EDGE, which marks an entry into the 3G realm of 2 Mbps cellular bandwidth.

While these standards will take time to stabilise and reach the user, there will be a consolidation in technologies where standards such as Wideband Code Division Multiple Access (WCDMA) will unify cellular communication standards, making data, voice, videoconferencing and multimedia messaging a part of everyday life.

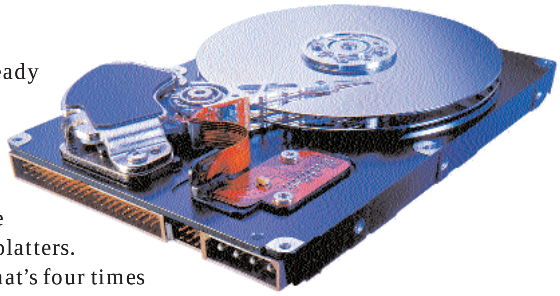


Processing powerhouses

Pulsing at the very core of our systems, processors have grown in strength to become massive number-crunching devices. Not too long ago, the prospect of having a computer that could allow you to create and edit your own video on your home computer was unheard of. But today's processors are more than able to handle power-hungry applications such as real-time audio and video processing, and bleeding-edge games with smoother levels of multitasking.

The year 2003 will see a slew of processors armed with higher clock speeds, advanced fabrication processes and newer technologies to tackle new-age applications with greater ease. AMD and Intel are going to herald 64-bit computing with their new processors aimed at the desktop and server segments. Aided by technologies such as Banias and better fabrication processes such as 0.09-micron technology, manufacturers will be able to pack in more transistors into processors, resulting in greater features and capability and higher integration of components. Besides the desktop computers, advances in processor fabrication will be reflected in handheld devices too—with clock speeds of up to 400 MHz, handheld processors will break new ground in portable power without sacrificing on battery life.

Storage unbound



The storage field has seen a lot of excitement in the past year. Hard disks already offer as much as 300 GB capacity and the bar for even entry-level drive capacities has been raised—the end of 2002 saw 40 GB drives paving the way for 60 GB ones. In 2003, advancements in fabrication technology will allow for even larger capacities to be integrated into the same form factor and price range. New technologies such as IBM's Pixie Dust allow the fabrication of devices where smaller magnetic domains can be reliably created and retained on the hard disk platters. This allows the disks to have storage densities of up to 100 billion bits per inch—that's four times the present capacities! You can look forward to hard disk drives with capacities beginning at 400 GB and notebook drives with up to 200 GB capacity!

On the optical drive front, Blue Ray is all set to redefine optical storage where standard 8 cm optical disks will boast of a 27 GB capacity. With many manufacturers already equipped to produce these drives, the only await is for the standards to support these technologies to be finalised. Another promising technology that is making its present felt is Recordable DVD drives. Given its superior storage capability and falling prices of hardware and media, this will become very popular.